

Module 06 – Transshipment Problem

Model Formulation

Decision Variables: the amount of goods shipped from node i to node j (0,1,2,3,4,5,6,7,8)

Objective Function: Minimize transportation Cost (the amount shipped $\times$ unit costs)

$$\min Z = 34x_{07} + 26x_{13} + 24x_{13} + 37x_{37} + 35x_{14} + 41x_{24} + 47x_{26} + 46x_{48} + 35x_{47} + 27x_{63} + 41x_{74} + 26x_{78} + 34x_{73} + 39x_{85} + 34x_{87}$$

Constraints: Supply (each supply node can only ship up to available supply) and Demand (each demand node must satisfy demand)

$$x_{07} \leq 300$$

$$x_{13} + x_{16} + x_{14} \leq 342$$

$$x_{24} + x_{26} \leq 256$$

$$x_{13} + x_{63} + x_{73} = 123$$

$$x_{14} + x_{24} + x_{74} = 184$$

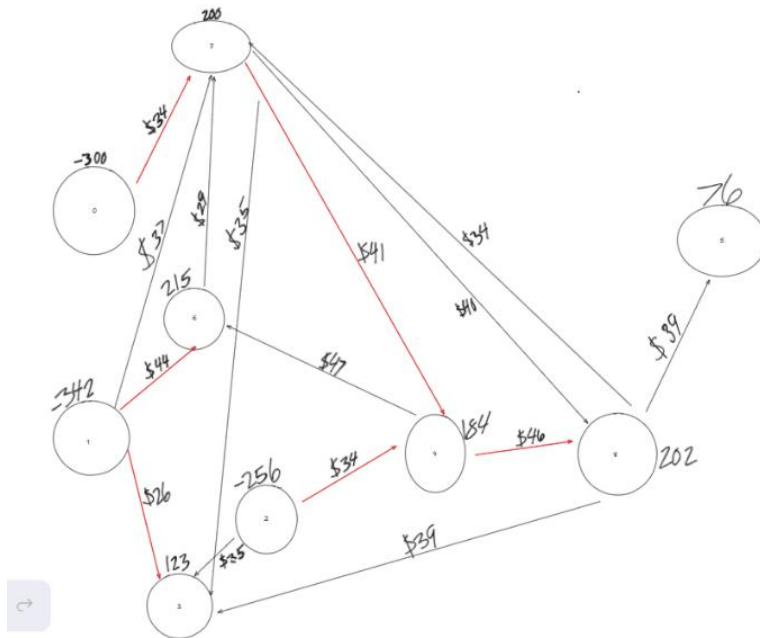
$$x_{85} = 76$$

$$x_{16} + x_{26} = 215$$

$$x_{07} + x_{37} + x_{47} + x_{87} = 200$$

$$x_{48} + x_{78} = 202$$

Model Optimized for Minimal Transportation Cost



Nodes	Inflow	Outflow	Net Flow	Supply/Demand
0 Butter Pecan Bluff	0	300	-300	-300
1 Fruity Gusher Geyser	0	342	-342	-342
2 Ginger Snap Garden	0	256	-256	-256
3 Jelly River Delta	123	0	123	123
4 Pineapple Pop Paradise	360	176	184	184
5 Praline Park	0	0	0	76
6 Soda Pop Springs	215	0	215	215
7 Starburst Starlit Skies	300	100	200	200
8 Toblerone Tower	176	0	176	202

This model suggests that following the transportation plan (amount shipped from one node to another) will achieve a total minimal transportation cost of \$43,898

Model with Stipulation

Balance of Flow constraint ensures that for transshipment nodes, the total goods coming into the node equal the total goods going out. This is necessary because it guarantees that no excess or deficit is left at the transshipment node.

When you change the model to make the total supply greater than demand by adding 155 units to a source, solver cannot find a feasible solution. For the model to work again, constraints need to be changed as demand is greater than supply after adding 115 units. When you make the demand greater than supply, the transportation cost increases.

Transportation Cost->					\$43,898.00
Ship	From	To	Unit Cost		
300	0 Butter Pecan Bluff	7 Starburst Starlit Skies	34		
123	1 Fruity Gusher Geyser	3 Jelly River Delta	26		
215	1 Fruity Gusher Geyser	6 Soda Pop Springs	44		
0	3 Jelly River Delta	7 Starburst Starlit Skies	37		
4	1 Fruity Gusher Geyser	4 Pineapple Pop Paradise	35		
256	2 Ginger Snap Garden	4 Pineapple Pop Paradise	34		
0	2 Ginger Snap Garden	6 Soda Pop Springs	47		
176	4 Pineapple Pop Paradis	8 Toblerone Tower	46		
0	4 Pineapple Pop Paradis	7 Starburst Starlit Skies	29		
0	6 Soda Pop Springs	3 Jelly River Delta	35		
100	7 Starburst Starlit Skies	4 Pineapple Pop Paradise	41		
0	7 Starburst Starlit Skies	8 Toblerone Tower	40		
0	7 Starburst Starlit Skies	3 Jelly River Delta	39		
0	8 Toblerone Tower	5 Praline Park	39		
0	8 Toblerone Tower	7 Starburst Starlit Skies	34		

				Transportation Cost->		\$ 47,044.00									
Ship	From		To		Unit Cost		Nodes		Inflow	Outflow	Net Flow	Supply/Demand			
326	0 Butter Pecan Bluff		7 Starburst Starlit Skies		34		0 Butter Pecan Bluff		0	326	-326	-415		300	
123	1 Fruity Gusher Geyser		3 Jelly River Delta		26		1 Fruity Gusher Geyser		0	342	-342	-342		342	
215	1 Fruity Gusher Geyser		6 Soda Pop Springs		44		2 Ginger Snap Garden		0	256	-256	-256		256	
0	3 Jelly River Delta		7 Starburst Starlit Skies		37		3 Jelly River Delta		123	0	123	123			
4	1 Fruity Gusher Geyser		4 Pineapple Pop Paradise		35		4 Pineapple Pop Paradise		386	202	184	184			
256	2 Ginger Snap Garden		4 Pineapple Pop Paradise		34		5 Praline Park		0	0	0	76			
0	2 Ginger Snap Garden		6 Soda Pop Springs		47		6 Soda Pop Springs		215	0	215	215			
202	4 Pineapple Pop Paradise		8 Toblerone Tower		46		7 Starburst Starlit Skies		326	126	200	200			
0	4 Pineapple Pop Paradise		7 Starburst Starlit Skies		29		8 Toblerone Tower		202	0	202	202			
0	6 Soda Pop Springs		3 Jelly River Delta		35										
126	7 Starburst Starlit Skies		4 Pineapple Pop Paradise		41										
0	7 Starburst Starlit Skies		8 Toblerone Tower		40										
0	7 Starburst Starlit Skies		3 Jelly River Delta		39										
0	8 Toblerone Tower		5 Praline Park		39										
0	8 Toblerone Tower		7 Starburst Starlit Skies		34										